Microsoft

From Local to Global: A Graph RAG Approach to Query-Focused Summarization

발표자 : 조수빈, 자연어처리팀

Contents

Introduction

- **Sensemaking** : 사람, 장소, 사건 간의 연결을 이해하고 그 흐름을 예측하여 효과적으로 행동하기 위한 지속적인 노력 (Klein et al., 2006a)
- **Mental Model** : 개인이 세상을 이해하고 해석하는데 사용하는 내적인 표현 또는 사고 프레임워크

- 현재 접근방법의 한계
  - **RAG (Retrieval-Augmented Generation)** :
    전체 text corpus 대상으로 하는 Global 질문에 대해서 대답하지 못함
  - **QFS (Query-Focused Summarization)** :
    방대한 양의 텍스트를 처리하기 어려움

→ **Graph RAG** : 사용자 질문의 범위와 처리할 텍스트의 양에 따라 유연하게 확장 가능

Related Work
- Graphs and LLMs

Fig. 1. Subset of the Knowledge Base generated using the REBEL model. The Knowledge Base is displayed in a graph format where entities are represented as nodes and relations are represented as edges.

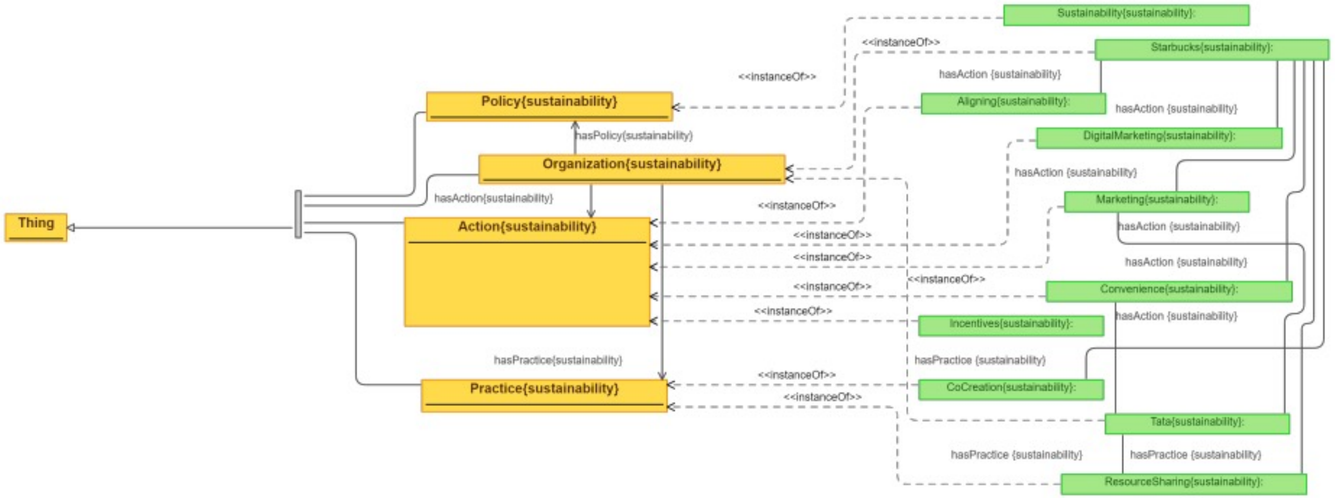


Fig. 4. Knowledge Base generated with ChatGPT for the second article. The identified concepts are represented as yellow rectangles, and the instances are represented with green rectangles.

Trajanoska, Milena, Riste Stojanov, and Dimitar Trajanov. "Enhancing knowledge graph construction using large language models." arXiv preprint arXiv:2305.04676 (2023).

Related Work
- Graphs and LLMs

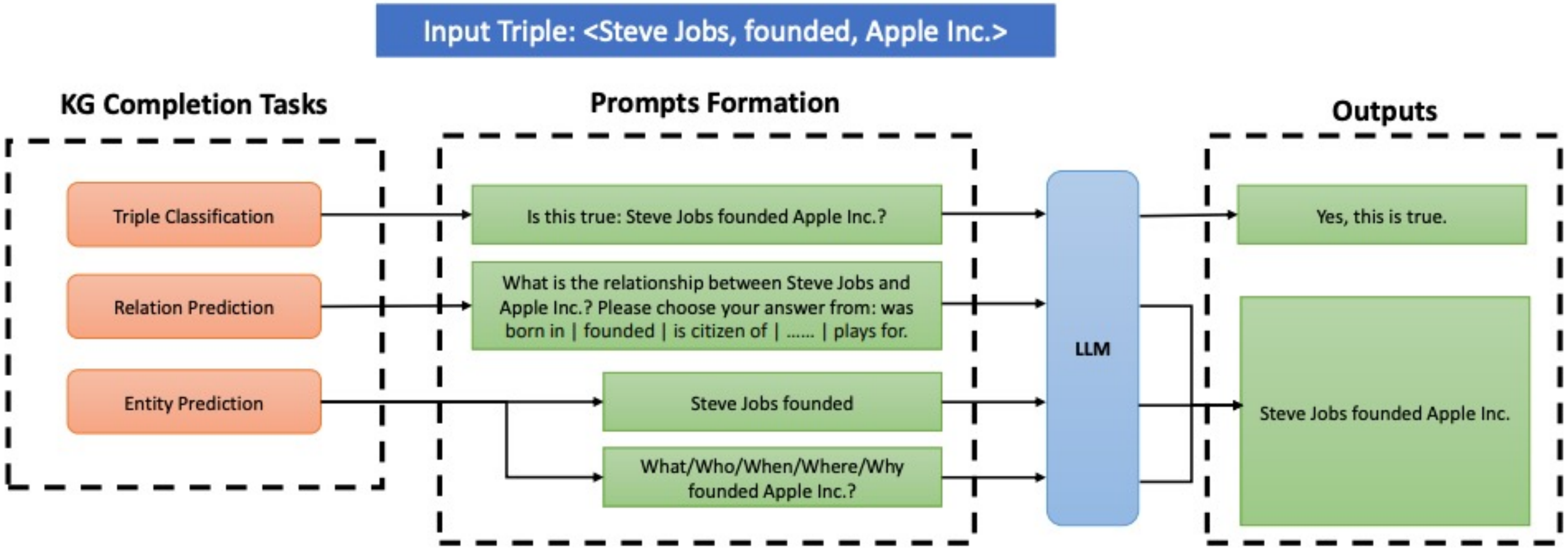


Figure 1: Illustrations of Large Langauge Models (LLMs) for Knowledge Graph (KG) Completion.

Yao, Liang, et al. "Exploring large language models for knowledge graph completion." arXiv preprint arXiv:2308.13916 (2023).

Related Work
- Graphs and LLMs

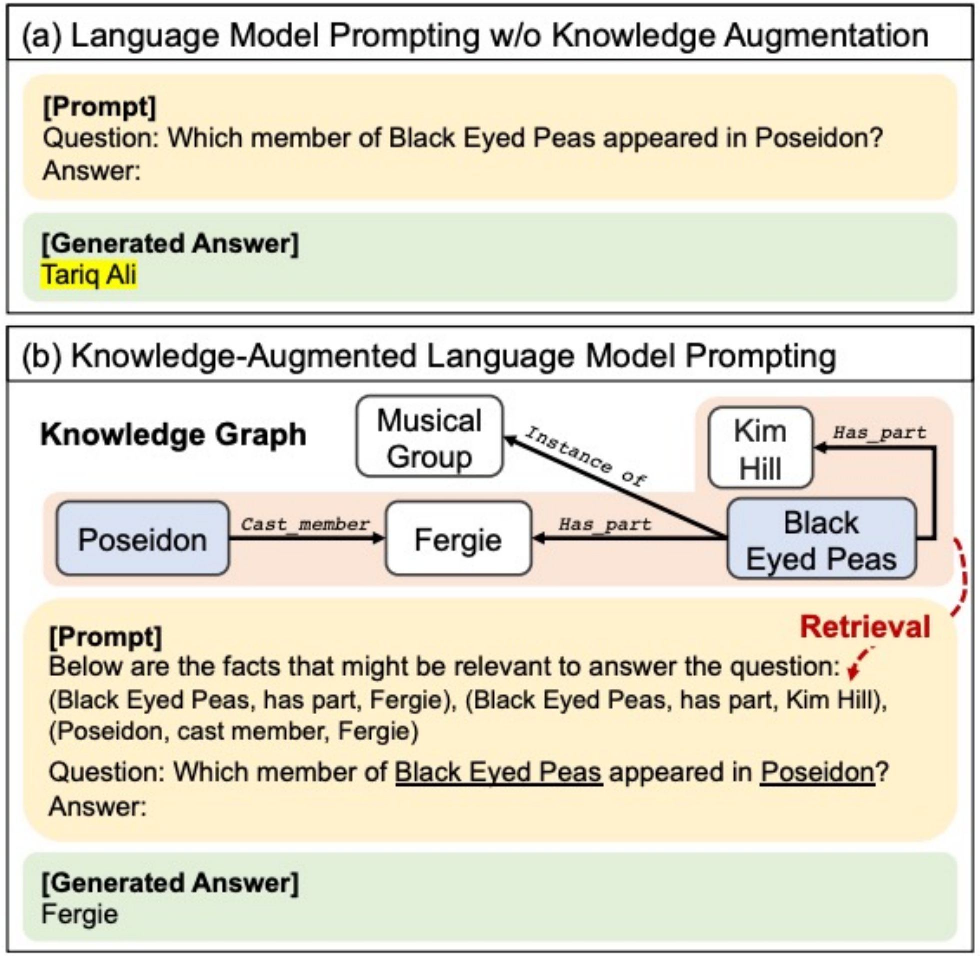


Figure 1: (a) For the input question in the prompt, the large language model, GPT-3 (Brown et al., 2020), can generate the answer based on its internal knowledge in parameters, but hallucinates it which is highlighted in yellow. (b) Our Knowledge-Augmented language model PrompTING (KAPING) framework first retrieves the relevant facts in the knowledge graph from the entities in the question, and then augments them to the prompt, to generate the factually correct answer.

Baek, Jinheon, Alham Fikri Aji, and Amir Saffari. "Knowledge-augmented language model prompting for zero-shot knowledge graph question answering." *arXiv preprint arXiv:2306.04136* (2023).

Graph RAG Pipeline

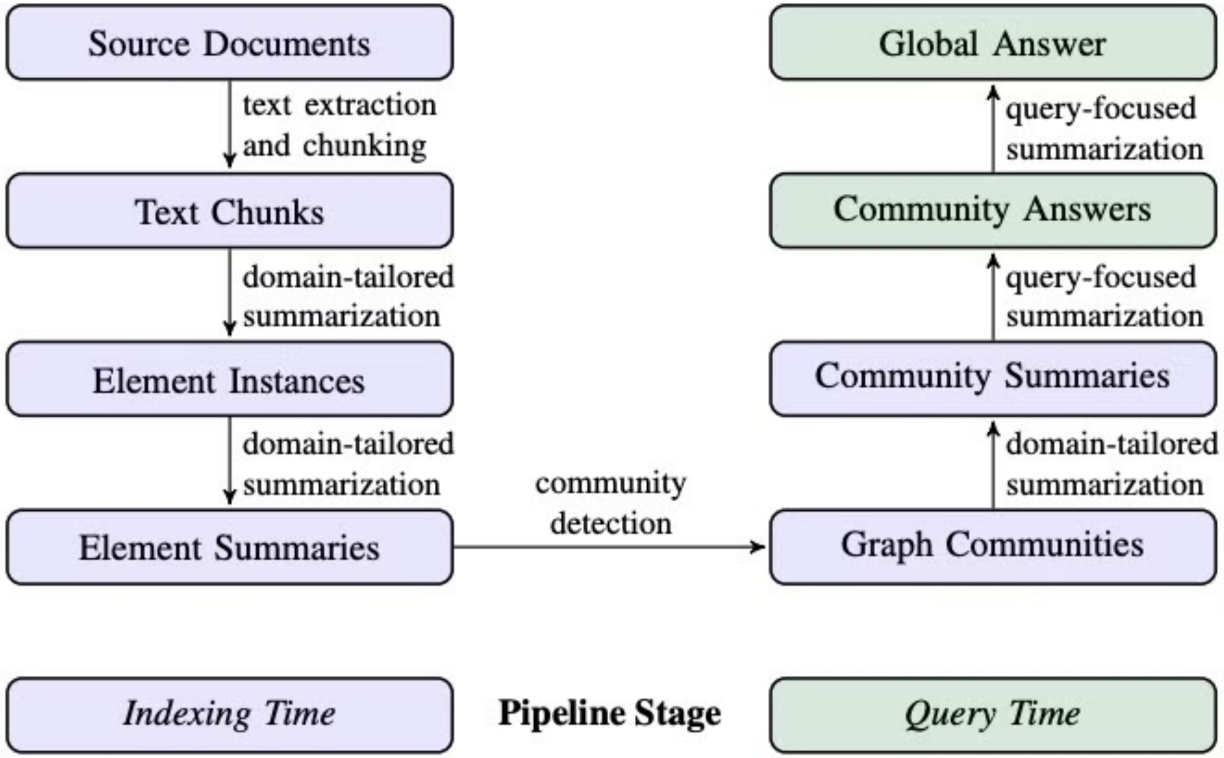


Figure 1: Graph RAG pipeline using an LLM-derived graph index of source document text. This index spans nodes (e.g., entities), edges (e.g., relationships), and covariates (e.g., claims) that have been detected, extracted, and summarized by LLM prompts tailored to the domain of the dataset. Community detection (e.g., Leiden, Traag et al., 2019) is used to partition the graph index into groups of elements (nodes, edges, covariates) that the LLM can summarize in parallel at both indexing time and query time. The "global answer" to a given query is produced using a final round of query-focused summarization over all community summaries reporting relevance to that query.

Graph RAG Pipeline

1. Source Documents → Text Chunks

- 원본 문서의 텍스트를 어떤 크기로 분할할지 결정

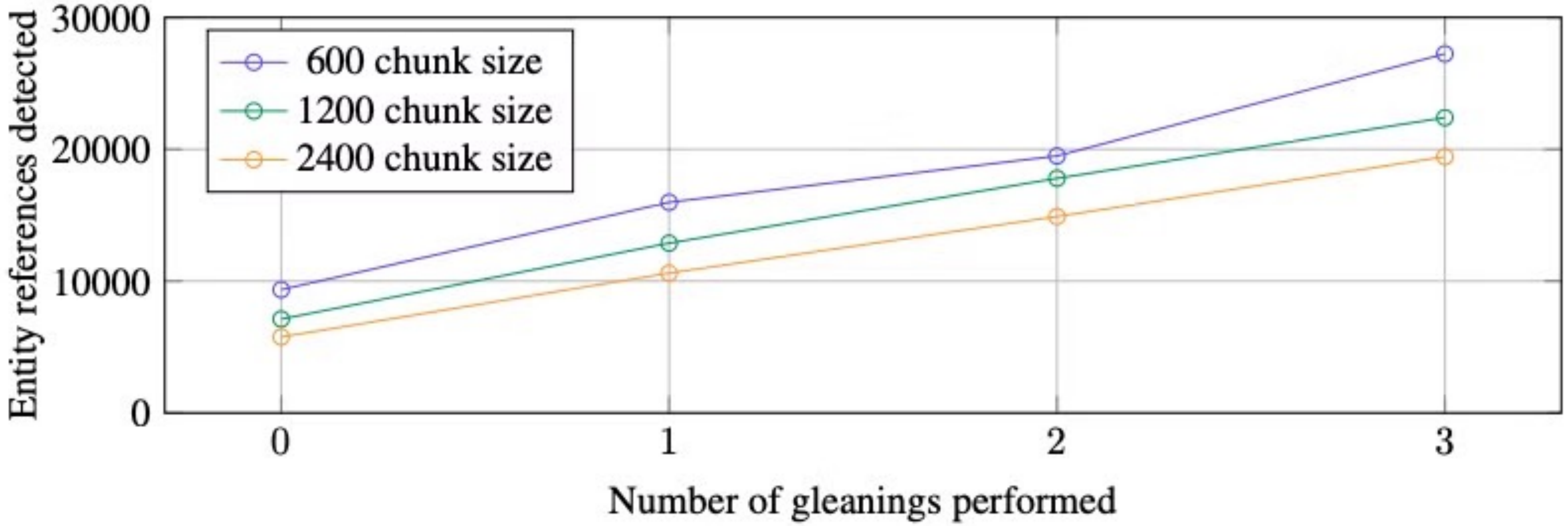


Figure 2: How the entity references detected in the HotPotQA dataset (Yang et al., 2018) varies with chunk size and gleanings for our generic entity extraction prompt with `gpt-4-turbo`.

Graph RAG Pipeline

2. Text Chunks → Element Instances

- 각 Text Chunk에서 Graph의 node와 edge 추출하는 단계
- 여러 라운드의 추가 탐색(gleanings) 과정을 사용하여 모든 entity를 추출

Graph RAG Pipeline

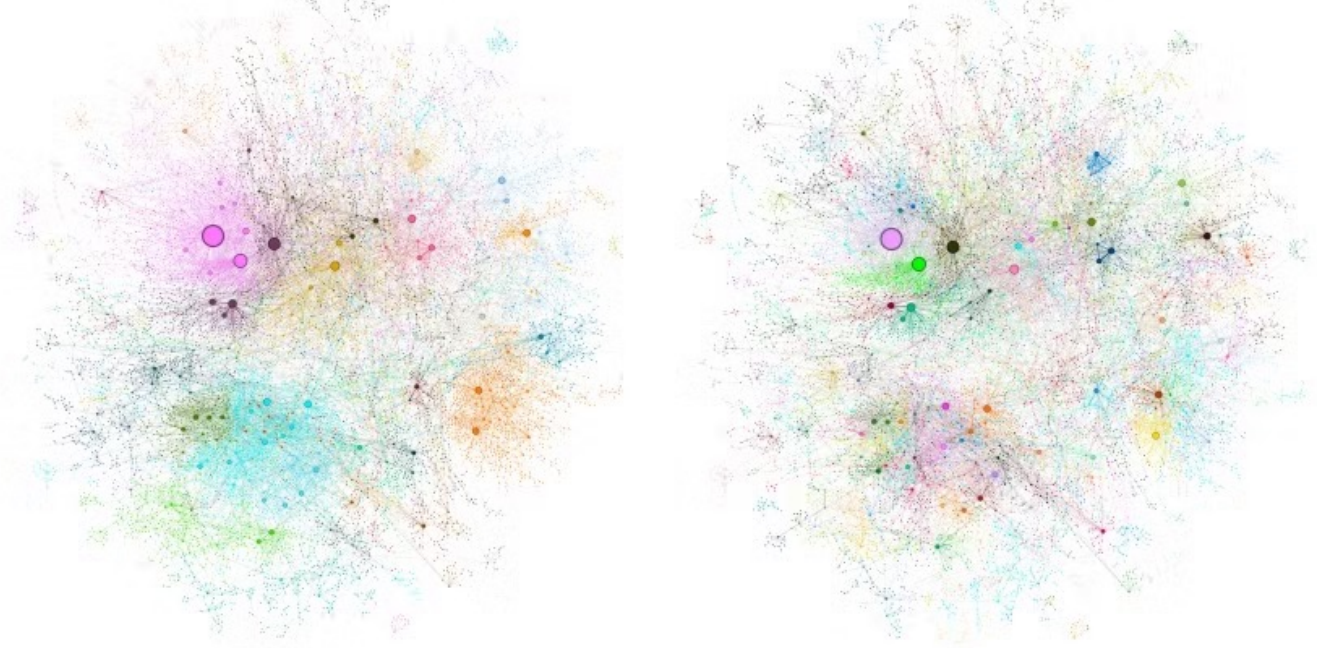
3. Element Instances → Element Summaries

- Element Instances를 각 Graph Element(entity node, relationship edge, claim covariate)에 대한 단일 설명 텍스트로 변환한 후,
- LLM의 요약 능력을 활용하여 일관성 있는 Element Summary 생성

Graph RAG Pipeline

4. Element Summaries → Graph Communities

- graph로 모델링 → Community Detection
- Leiden 알고리즘 사용



(a) Root communities at level 0 (b) Sub-communities at level 1

Figure 3: Graph communities detected using the Leiden algorithm (Traag et al., 2019) over the MultiHop-RAG (Tang and Yang, 2024) dataset as indexed. Circles represent entity nodes with size proportional to their degree. Node layout was performed via OpenORD (Martin et al., 2011) and Force Atlas 2 (Jacomy et al., 2014). Node colors represent entity communities, shown at two levels of hierarchical clustering: (a) Level 0, corresponding to the hierarchical partition with maximum modularity, and (b) Level 1, which reveals internal structure within these root-level communities.

Leiden 알고리즘

1. local moving of nodes
2. refinement of the partition
3. aggregation of the network

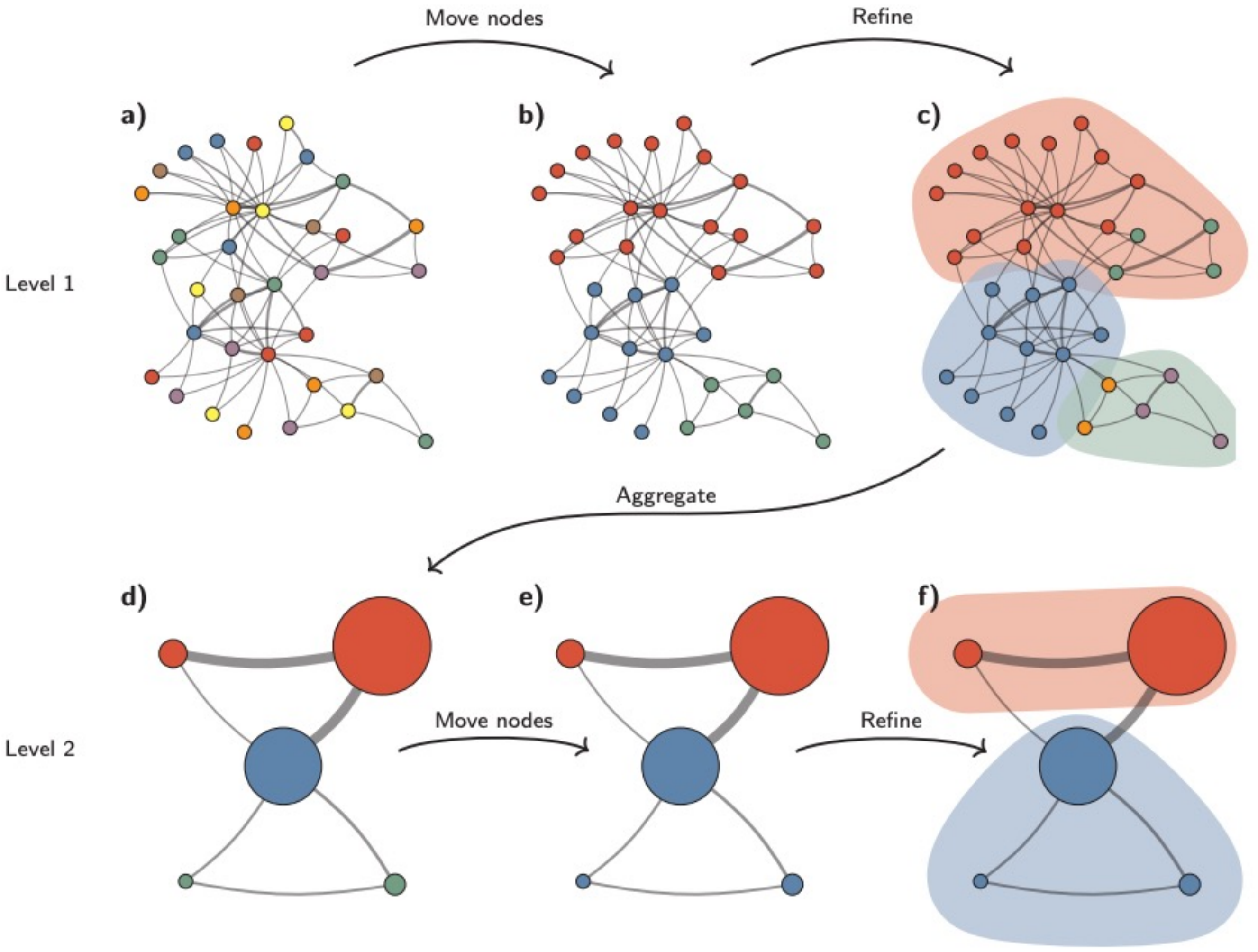


FIG. 3. **Leiden algorithm**. The Leiden algorithm starts from a singleton partition (a). The algorithm moves individual nodes from one community to another to find a partition (b), which is then refined (c). An aggregate network (d) is created based on the refined partition, using the non-refined partition to create an initial partition for the aggregate network. For example, the red community in (b) is refined into two subcommunities in (c), which after aggregation become two separate nodes in (d), both belonging to the same community. The algorithm then moves individual nodes in the aggregate network (e). In this case, refinement does not change the partition (f). These steps are repeated until no further improvements can be made.

Traag, V. A., L. Waltman, and N. J. Van Eck. "From Louvain to Leiden: guaranteeing well-connected communities. Sci. Rep. 9, 5233." (2019)..

Graph RAG Pipeline

5. Graph Communities → Community Summaries

- Leiden 계층의 각 커뮤니티에 대한 요약을 생성
- Leaf-level community → higher-level community

Graph RAG Pipeline

6. Community Summaries → Community Answers → Global Answer

1. **Prepare community summaries** :
   community summary를 일정 크기의 chunk로 나눔

2. **Map community answers** :
   각 chunk에 대해 병렬로 community answer 생성

3. **Reduce to global answer** :
   유용성 점수에 따라 정렬하고 컨텍스트 윈도우에 추가하여 최종 global answer 생성

Evaluation
1. Datasets & Queries

- Datasets : Podcast transcripts, News articles
- LLM을 활용하여 전체 corpus의 이해가 필요한 질문을 생성
- 데이터셋 당 125개의 질문
  5(user)x5(task)x5(question)

Dataset	Example activity framing and generation of global sensemaking questions
Podcast transcripts	*User*: A tech journalist looking for insights and trends in the tech industry *Task*: Understanding how tech leaders view the role of policy and regulation *Questions*: 1. Which episodes deal primarily with tech policy and government regulation? 2. How do guests perceive the impact of privacy laws on technology development? 3. Do any guests discuss the balance between innovation and ethical considerations? 4. What are the suggested changes to current policies mentioned by the guests? 5. Are collaborations between tech companies and governments discussed and how?
News articles	*User*: Educator incorporating current affairs into curricula *Task*: Teaching about health and wellness *Questions*: 1. What current topics in health can be integrated into health education curricula? 2. How do news articles address the concepts of preventive medicine and wellness? 3. Are there examples of health articles that contradict each other, and if so, why? 4. What insights can be gleaned about public health priorities based on news coverage? 5. How can educators use the dataset to highlight the importance of health literacy?

Table 1: Examples of potential users, tasks, and questions generated by the LLM based on short descriptions of the target datasets. Questions target global understanding rather than specific details.

Evaluation

2. Conditions

- Graph RAG (graph communities level - C0, C1, C2, C3)
- Text Summarization Method (TS)
- Semantic Search RAG Approach (SS)

Evaluation

3. Metrics

- **Comprehensiveness** :
  답변이 질문의 모든 측면과 세부 사항을 다루기 위해 얼마나 많은 세부 정보를 제공하는가?
- **Diversity** :
  답변이 질문에 대한 다양한 관점과 통찰력을 제공하는 데 얼마나 다양하고 풍부한가?
- **Empowerment** :
  답변이 독자가 주제를 이해하고 정보에 기반한 판단을 내리는 데 얼마나 도움이 되는가?
- **Directness** :
  답변이 질문을 얼마나 구체적이고 명확하게 다루는가?

- 질문, 4개의 목표 Metrics, 한 쌍의 답변 → **LLM** → 더 나은 답변 평가와 그 이유
- 각 비교를 5번 실행하고 평균 점수를 사용

Evaluation

3. Metrics

Question	Which public figures are repeatedly mentioned across various entertainment articles?
Graph RAG	**Overview of Prominent Public Figures in Entertainment** The entertainment industry is vast and diverse, encompassing film, television, music, sports, and digital media. Certain public figures stand out due to their significant contributions and influence across these sectors. The following summary highlights key individuals who are repeatedly mentioned in various entertainment articles, reflecting their impact and presence within the industry. **Actors and Directors** [...] **Public Figures in Controversy** [...] **Musicians and Executives** [...] **Athletes and Coaches** [...] **Influencers and Entrepreneurs** [...] The repeated mention of these figures in entertainment articles signifies their ongoing relevance and the public's interest in their work. Their influence spans across various aspects of entertainment, from shaping cultural narratives in film and television to driving trends in music and digital media. These individuals not only contribute to their respective fields but also influence the broader cultural landscape, often becoming central figures in social discussions and public discourse.
Naïve RAG	Public figures who are repeatedly mentioned across various entertainment articles include Taylor Swift, Travis Kelce, Britney Spears, and Justin Timberlake. These individuals have been highlighted for various reasons, ranging from their professional achievements to their personal lives. **Taylor Swift** [...] **Travis Kelce** [...] **Britney Spears** [...] **Justin Timberlake** [...] These figures are frequently covered due to their high-profile status and the public's interest in their careers and personal lives. Their activities, whether in music, sports, or personal relationships, have significant cultural and economic impacts, as evidenced by the media coverage and public reactions.
LLM Decision	**Comprehensiveness: Winner=1 (Graph RAG)** Answer 1 is better because it provides a more comprehensive and detailed list of public figures from a wider range of entertainment sectors, including film, television, music, sports, gaming, and digital media. It also includes specific examples of their contributions and the impact they have on their respective fields, as well as mentions of controversies and their implications. Answer 2, while detailed in its coverage of a few individuals, is limited to a smaller number of public figures and focuses primarily on their personal lives and relationships rather than a broad spectrum of their professional influence across the entertainment industry.

Evaluation

3. Metrics

	Diversity: Winner=1 (Graph RAG) Answer 1 is better because it provides a more varied and rich response by covering a wide range of public figures from different sectors of the entertainment industry, including film, television, music, sports, gaming, and digital media. It offers insights into the contributions and influence of these figures, as well as controversies and their impact on public discourse. The answer also cites specific data sources for each mentioned figure, indicating a diverse range of evidence to support the claims. In contrast, Answer 2 focuses on a smaller group of public figures, primarily from the music industry and sports, and relies heavily on a single source for data, which makes it less diverse in perspectives and insights. **Empowerment: Winner=1 (Graph RAG)** Answer 1 is better because it provides a comprehensive and structured overview of public figures across various sectors of the entertainment industry, including film, television, music, sports, and digital media. It lists multiple individuals, providing specific examples of their contributions and the context in which they are mentioned in entertainment articles, along with references to data reports for each claim. This approach helps the reader understand the breadth of the topic and make informed judgments without being misled. In contrast, Answer 2 focuses on a smaller group of public figures and primarily discusses their personal lives and relationships, which may not provide as broad an understanding of the topic. While Answer 2 also cites sources, it does not match the depth and variety of Answer 1. **Directness: Winner=2 (Naïve RAG)** Answer 2 is better because it directly lists specific public figures who are repeatedly mentioned across various entertainment articles, such as Taylor Swift, Travis Kelce, Britney Spears, and Justin Timberlake, and provides concise explanations for their frequent mentions. Answer 1, while comprehensive, includes a lot of detailed information about various figures in different sectors of entertainment, which, while informative, does not directly answer the question with the same level of conciseness and specificity as Answer 2.

Table 2: Example question for the News article dataset, with generated answers from Graph RAG (C2) and Naïve RAG, as well as LLM-generated assessments.

Evaluation

4. Results

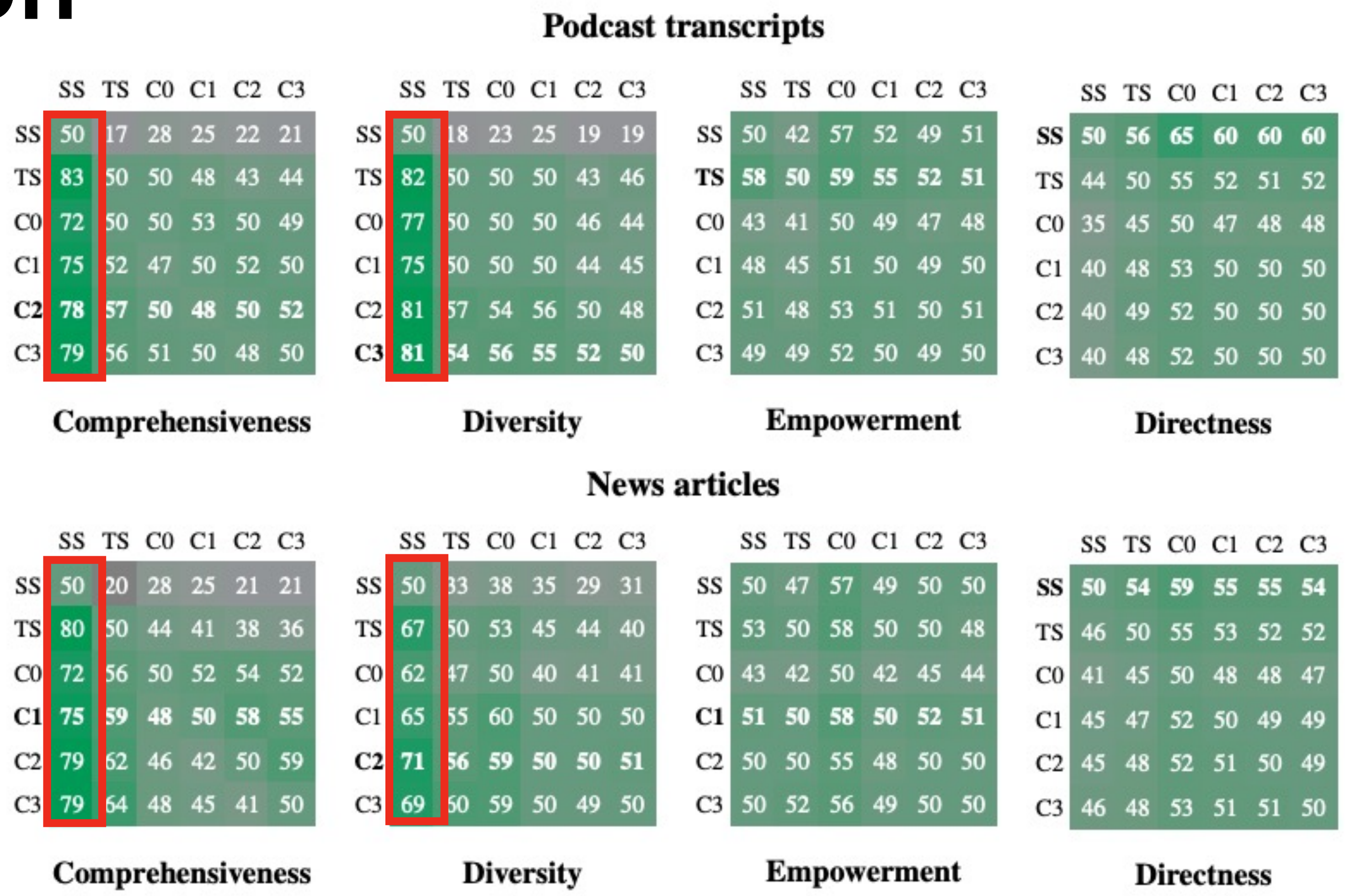


Figure 4: Head-to-head win rate percentages of (row condition) over (column condition) across two datasets, four metrics, and 125 questions per comparison (each repeated five times and averaged). The overall winner per dataset and metric is shown in bold. Self-win rates were not computed but are shown as the expected 50% for reference. All Graph RAG conditions outperformed naïve RAG on comprehensiveness and diversity. Conditions C1-C3 also showed slight improvements in answer comprehensiveness and diversity over TS (global text summarization without a graph index).

Evaluation
4. Results

	Podcast Transcripts					News Articles				
	C0	**C1**	**C2**	**C3**	**TS**	**C0**	**C1**	**C2**	**C3**	**TS**
Units	34	367	969	1310	1669	55	555	1797	2142	3197
Tokens	26657	225756	565720	746100	1014611	39770	352641	980898	1140266	1707694
% Max	2.6	22.2	55.8	73.5	100	2.3	20.7	57.4	66.8	100

Table 3: Number of context units (community summaries for **C0-C3** and text chunks for **TS**), corresponding token counts, and percentage of the maximum token count. Map-reduce summarization of source texts is the most resource-intensive approach requiring the highest number of context tokens. Root-level community summaries (**C0**) require dramatically fewer tokens per query (9x-43x).

Conclusion

- Knowledge Graph, RAG, QFS 결합한 global Graph RAG 접근법 제시
- 다른 전역적 방법들과 비슷한 성능을 내면서도 토큰 비용 절감 가능